

Drug Delivery: Engineering and Principles

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Drug Delivery: Principles and Applications

- BE 210 (AUG) 3:0
- TA: Kiran Rao U (kiranu@iisc.ac.in) and Subrahmanyam B R (subrahmanya6@iisc.ac.in)
- Generally, I would be available after classes
- Do drop me an email if you plan to come to my office to meet.
- Course Eligibility:
 - All graduate students
 - Undergraduates who have completed their 2nd year and counts as assorted credits

Drug Delivery: Principles and Applications

- Prerequisites: None. Some background in biology is preferred but not required
- All lecture videos are available freely (and transcribed) through NPTEL platform
 - <https://nptel.ac.in/courses/102/108/102108077/>
- References (not required):
 - Drug Delivery: Engineering Principles for Drug Therapy by Mark Saltzman
 - Drug Delivery: Fundamentals and Applications, Anya M. Hillery and Kinam Park, 2nd Edition, CRC Press, 2016

Grading Scheme

- Class participation: 28 marks
 - Announced quizzes (2 to 4): 10-15 marks
 - If you missed a quiz due to medical or other valid reason, then either you can be given the average of the quiz for the whole class, or I can take your viva separately
 - Class participation: 10-15 marks
 - Attendance will be taken in a few classes (quiz and group projects).
- Punctual
- Exam (end of November): 30 marks
- Group Project (presentation by mid Nov): 45 marks
 - May require a separate 2-3 h session outside of the class schedule
- Grading: Will curve

Course Outline

- Introduce concepts of drug delivery for medical challenges
- Pharmacokinetics: Bioavailability, Elimination, Therapeutic index
- Prodrugs, Controlled release
- Polymers: Synthesis, properties, characterization, crystallinity and amorphousness
- Biopolymers: Natural and Synthetic, biocompatibility, Biodegradation, commonly used biopolymers
- Polymer-Drug conjugates, PEGylation
- Diffusion controlled systems, Ficks laws, Reservoir systems, Non-erodible matrix systems, Bio-erodible Systems

Course Outline

- Hydrogels: Physical or chemical, in-situ crosslinking
- Nano and Micro-particles: Dendrimers, Liposomes, Micelles
- Metal and polymeric particles, effect of particle shape, charge and elasticity
- Protein Adsorption and tissue engineering, Drug delivery in tissue engineering
- Implant associated infections, Route specific delivery: Oral, Subcutaneous, Intramuscular, transdermal, inhalation, intravenous
- Vaccines, Cancer vaccines, Cell and gene delivery, Smart responsive drug delivery, Targeted drug delivery, Nanotoxicology and market translation

Course Outline

- Key papers will be discussed throughout the course
- Group project: Propose a project with drug delivery application for an existing medical need
 - Clear need
 - Survey of latest literature
 - Strategy
 - Pros and cons

Drug Delivery

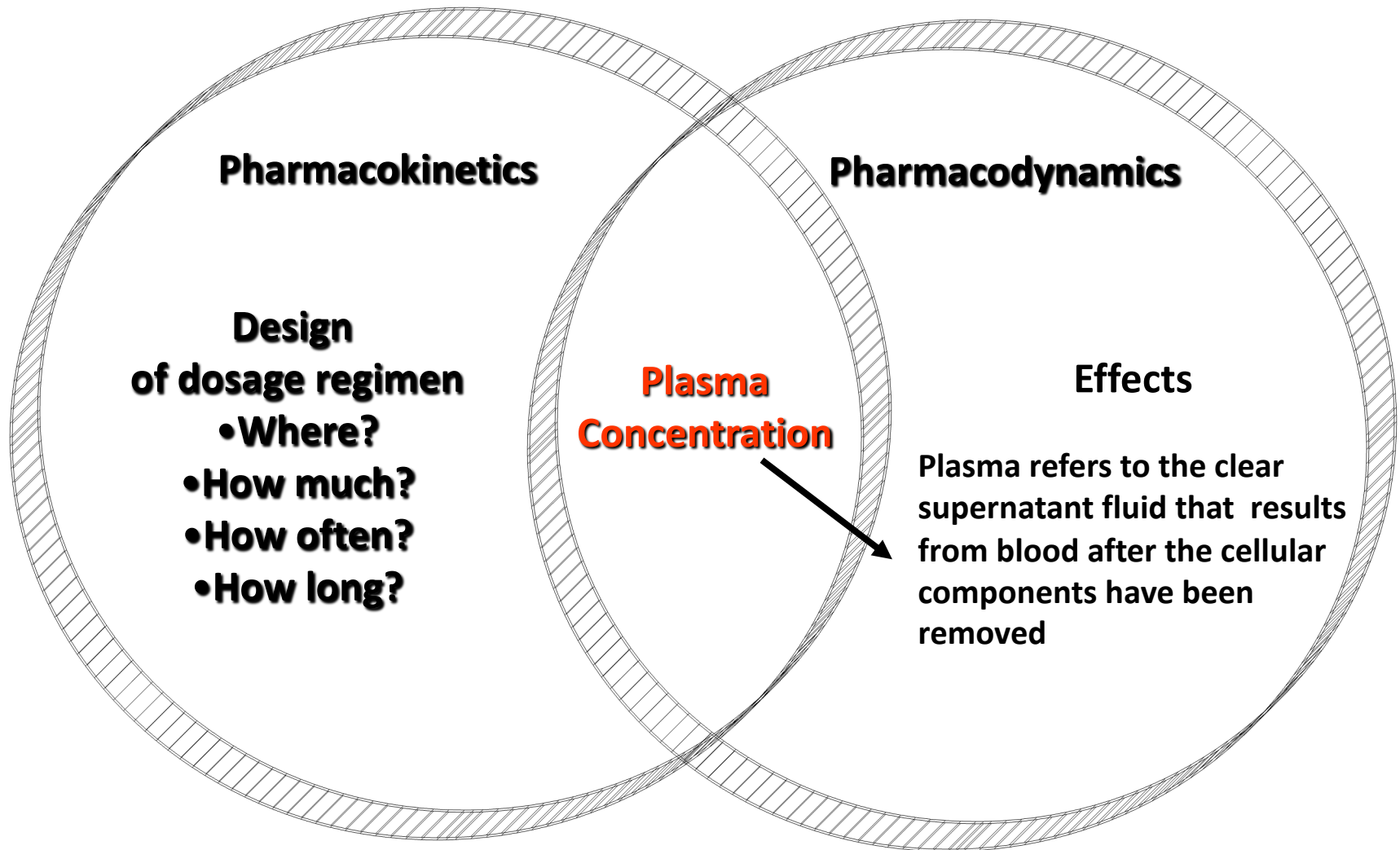
- The appropriate administration of drugs through various routes in the body for the purpose of improving health
 - All diseases need drugs to be delivered
 - Highly interdisciplinary
 - Biology
 - Physiology
 - Materials
 - Engineering
 - It has recently evolved to take into consideration
 - Drug physico-chemical properties
 - Body effects and interactions
 - Improvement of drug effect
- Eventual goal is patient comfort and well being

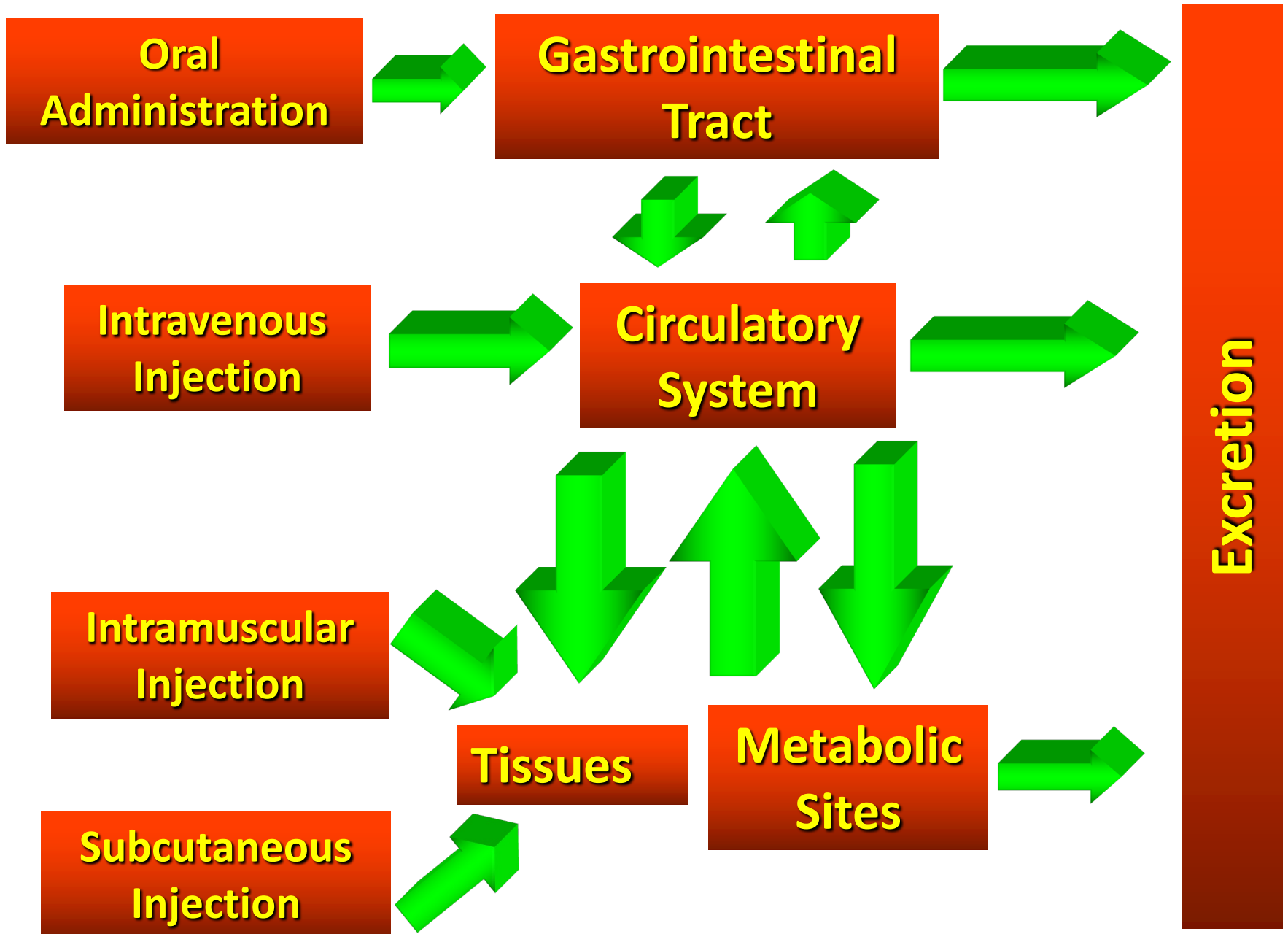


Magnitude of Drug Response

- Depends upon concentration achieved at the site of action
 - Dosage
 - Extent of absorption
 - Distribution to the site
 - Rate/extent of elimination

Pharmacokinetics and Pharmacodynamics





Implications of PK and PD in Drug Delivery

- **The PK and PD of a drug may be affected when administered via different routes**
 - **Examples**
 - **Proteins – oral vs. intramuscular**
- **The PK and PD of a drug delineates its therapeutic window**
 - **Degree of absorption**
 - **Degree of elimination and/or metabolism**
 - **Example**
 - **Tetracycline (infection) – given 6 to 8 hours**
 - **Digoxin (cardiac failure)– given daily**

Factors affecting pharmacokinetics

Solubility

Partition coefficient

Ionization

Several others

Solubility

Drugs must be in solution (not precipitated) to interact with receptors/targets

Drugs have some degree of solubility in both aqueous and lipid compartments (PC)

Solubility is a function of:

- ionization

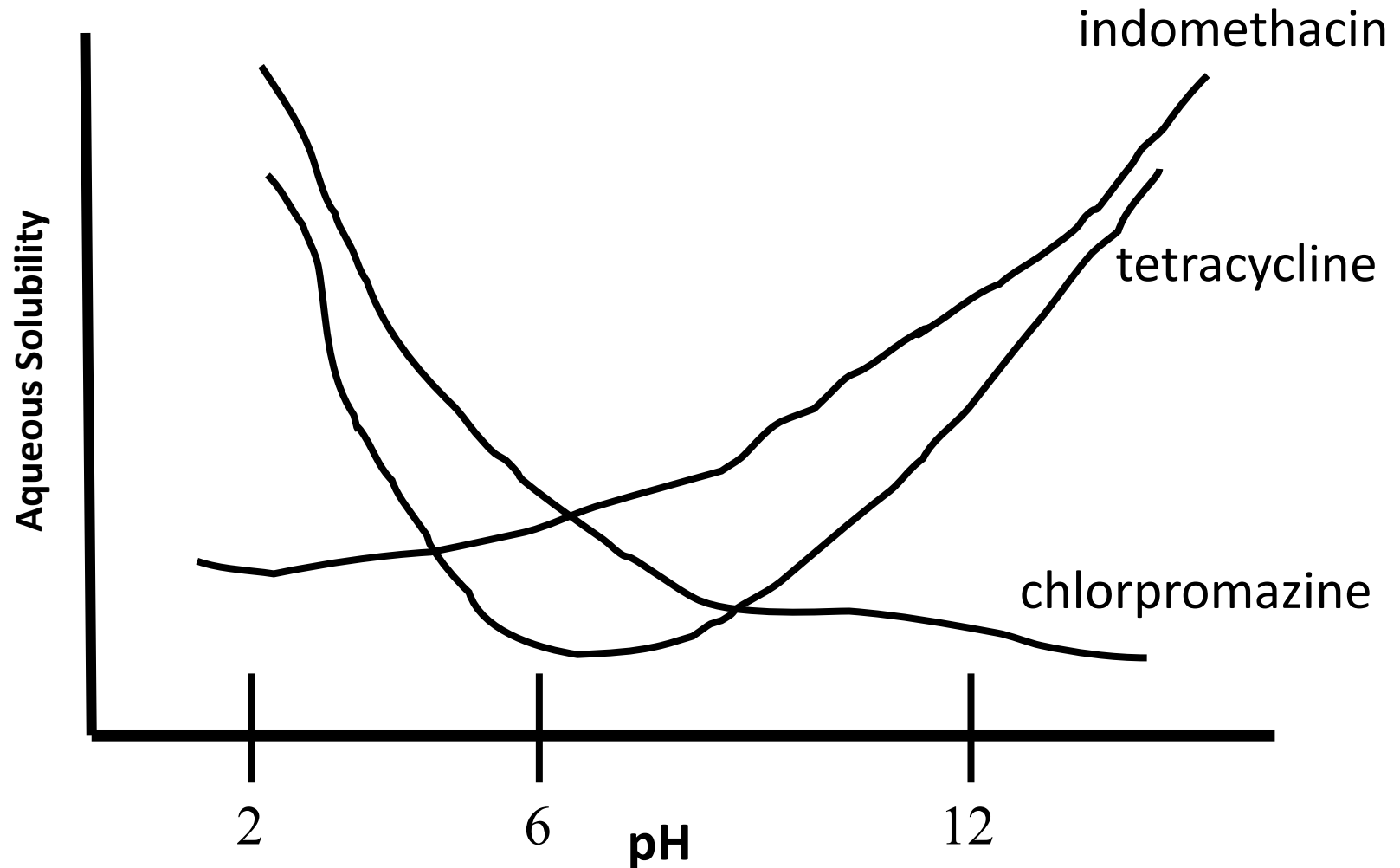
- molecular structure

- molecular weight

- electronic structure

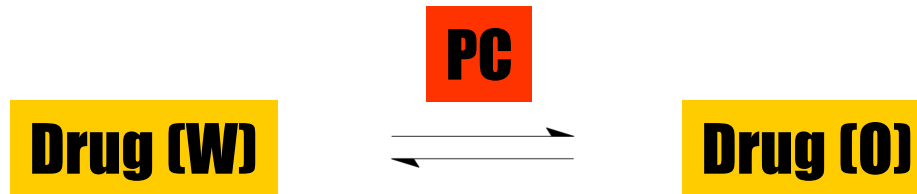
Solubility: Examples

Solubility will be affected by pH



Partition Coefficient

The ratio of the concentrations of a solute in two immiscible or slightly miscible liquids when it is in equilibrium across the interface between them



$$\text{PC} = \frac{[\text{drug}]_o}{[\text{drug}]_w}$$

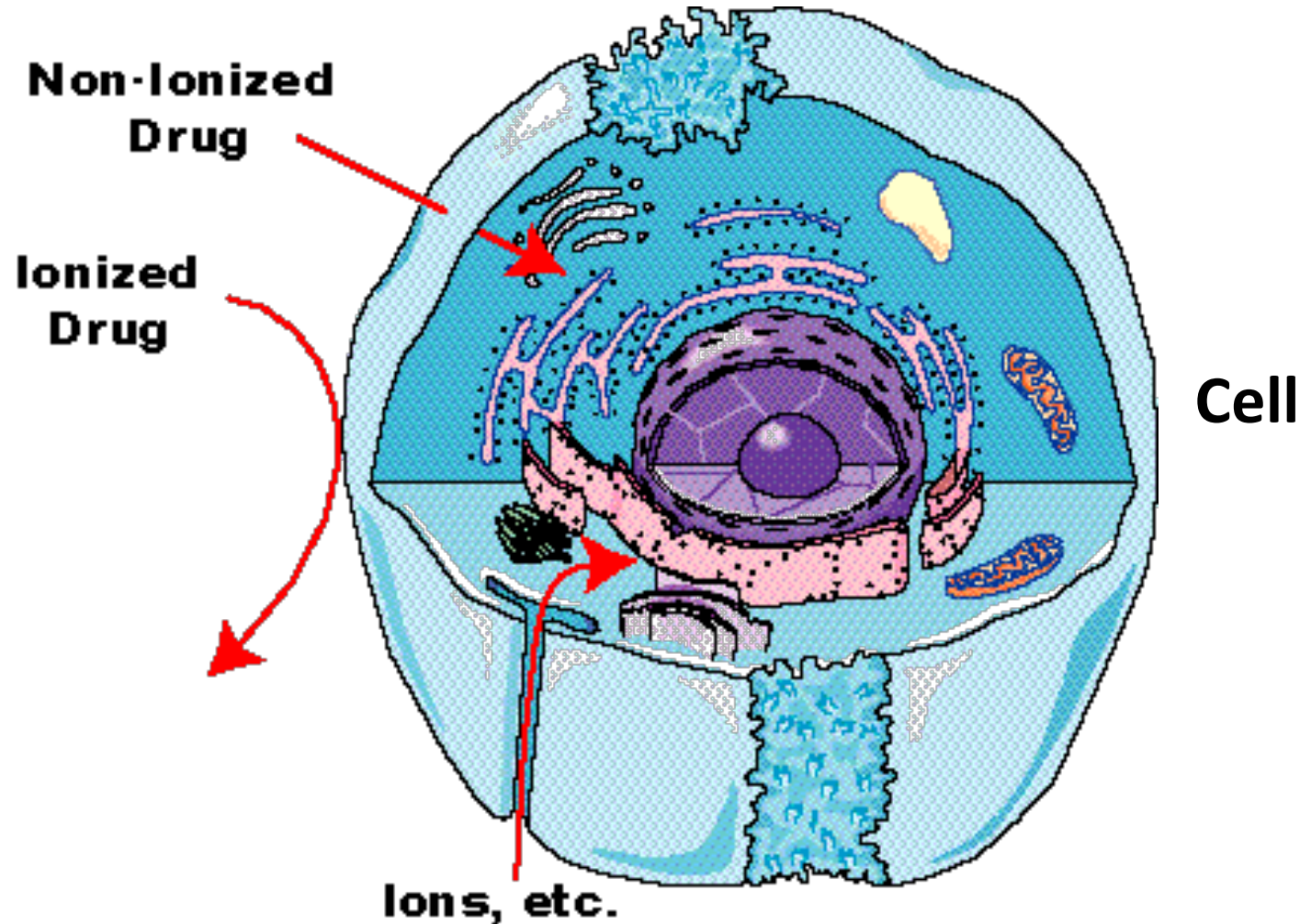
**Thus, PC
describes
the entire
drug.**

Lipid-Water Partition Coefficient

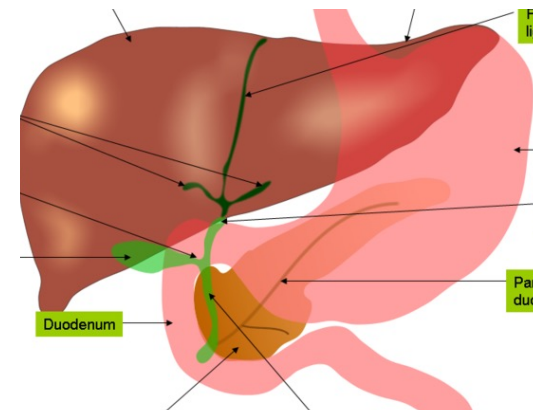
- The higher the lipid/water p.c. the greater the rate of transfer across the membrane
 -  polarity of a drug, by increasing ionization will  the lipid/ water p.c.
 -  polarity of a drug, suppression of ionization will  the lipid/ water p.c.

Drug Ionization

Charged molecules typically do not enter drug by passive diffusion

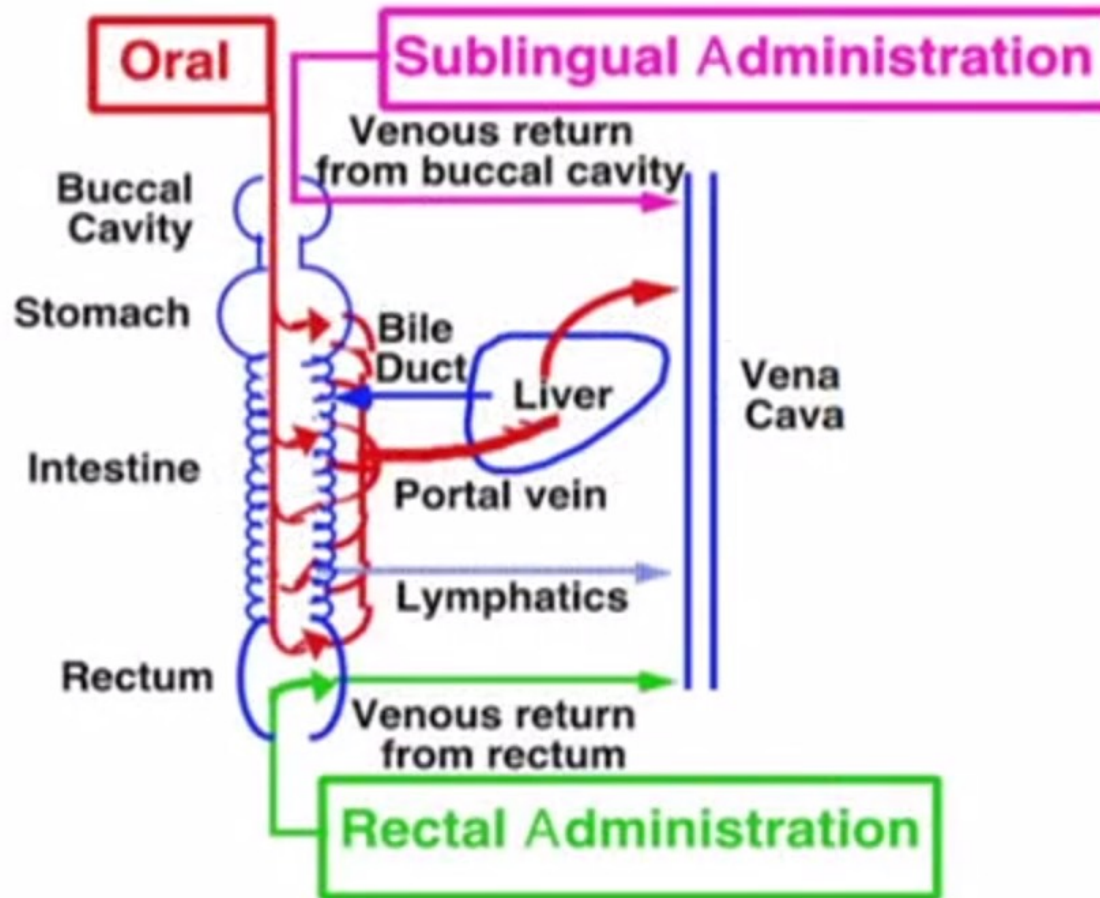


First-pass Effect: Oral Delivery



- Hepatic metabolism of a pharmacological agent
- Absorbed from the gut and delivered to the liver via the portal circulation
- The greater the first-pass effect, the less the agent will reach the systemic circulation
- Applicable when the agent is administered orally

ENTERAL ABSORPTION OF DRUGS



Sublingual route has lower barriers but also has low absorption

Rectal administration prevent first pass metabolism for more than 50% of the drug

Exercise

Let's see how we are doing

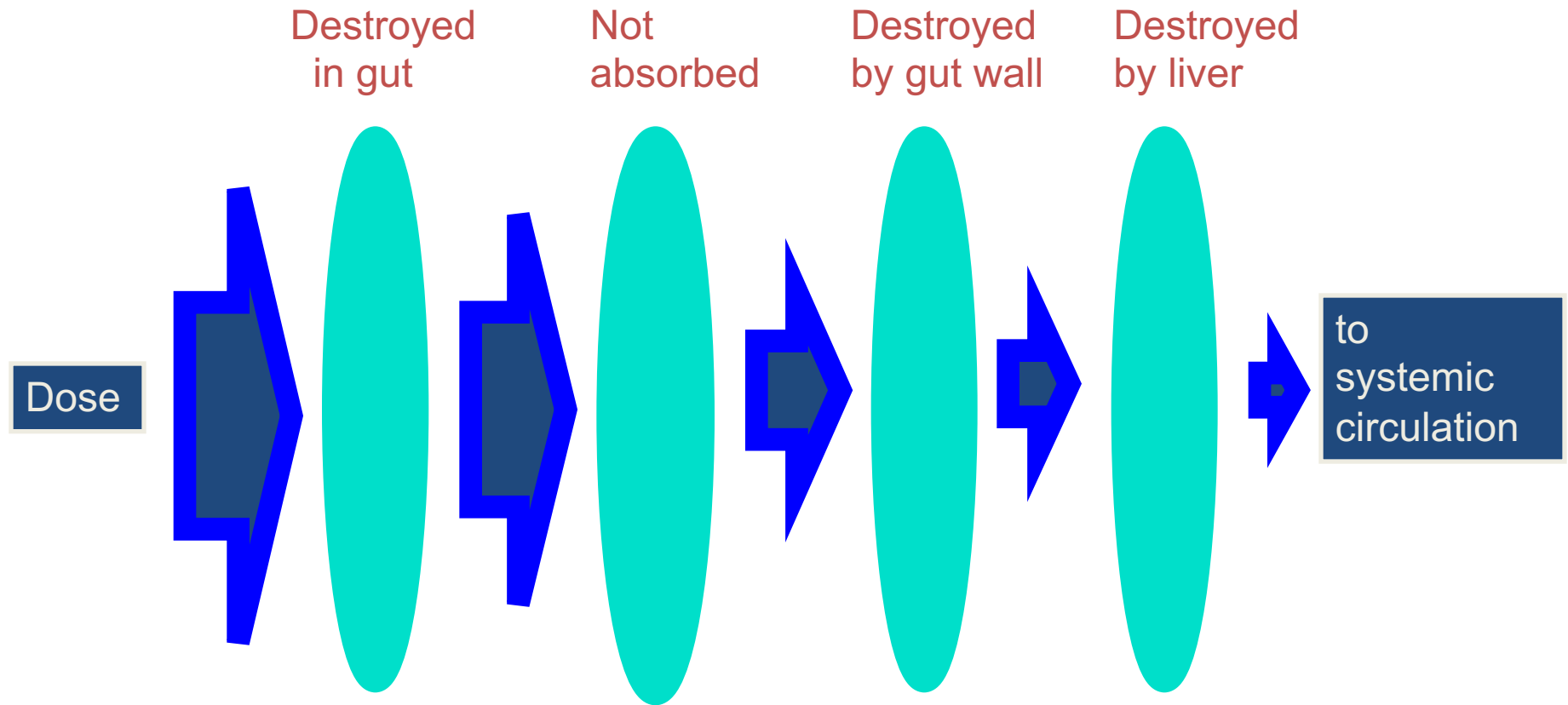
- The first pass effect refers to
 - a. The absorption of a drug from its site of administration
 - b. How the drug reaches its first site of action
 - c. How the amount of a drug can be reduced by metabolism before it reaches the systemic circulation.
 - d. How a drug is metabolized after it reaches the systemic circulation

Answer is c

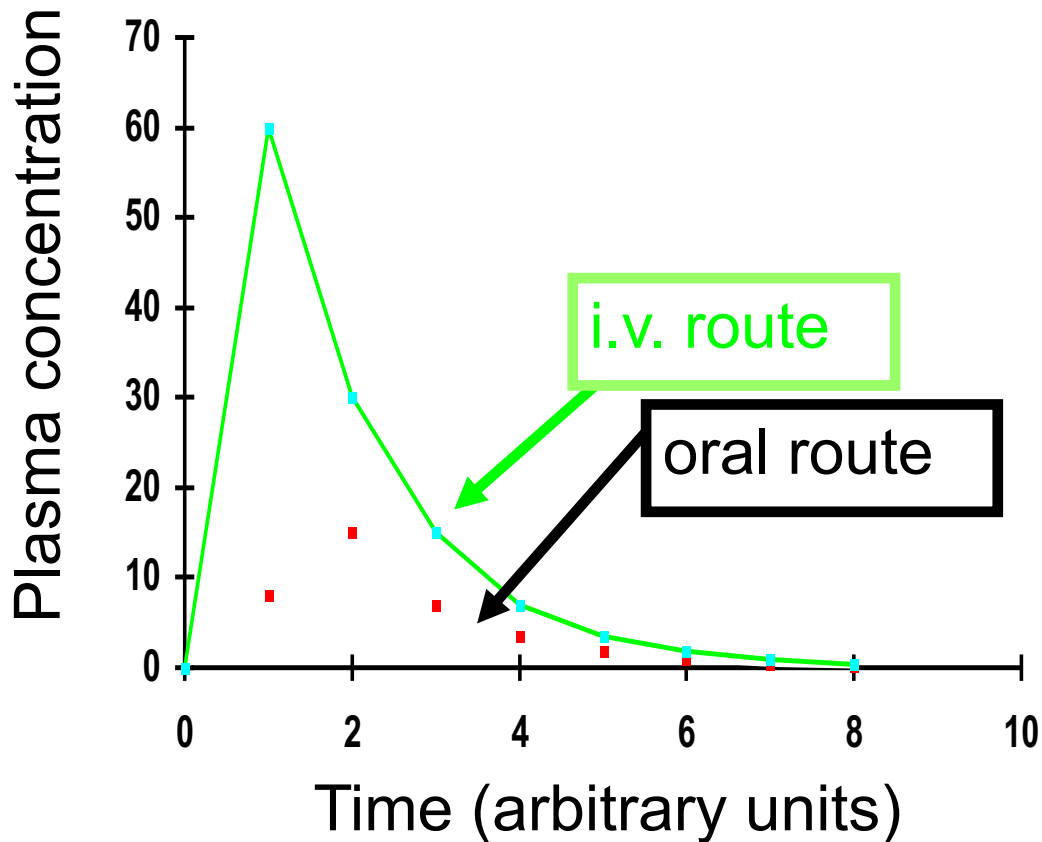
Bioavailability

- Essential to determine what dose will induce the desired therapeutic effect
- Different routes can lead different bioavailability
- 100% of the administered drug is bioavailable when delivered through IV route if it is soluble in aqueous medium
- Other routes may have different efficiencies of drug that are bioavailable even when administered at same dose
- Explains why the same dose may cause a therapeutic effect by one route but a toxic or no effect by another

Bioavailability: Oral Delivery



$$\text{Bioavailability} = \frac{(\text{AUC})_{\text{route}}}{(\text{AUC})_{\text{iv}}}$$



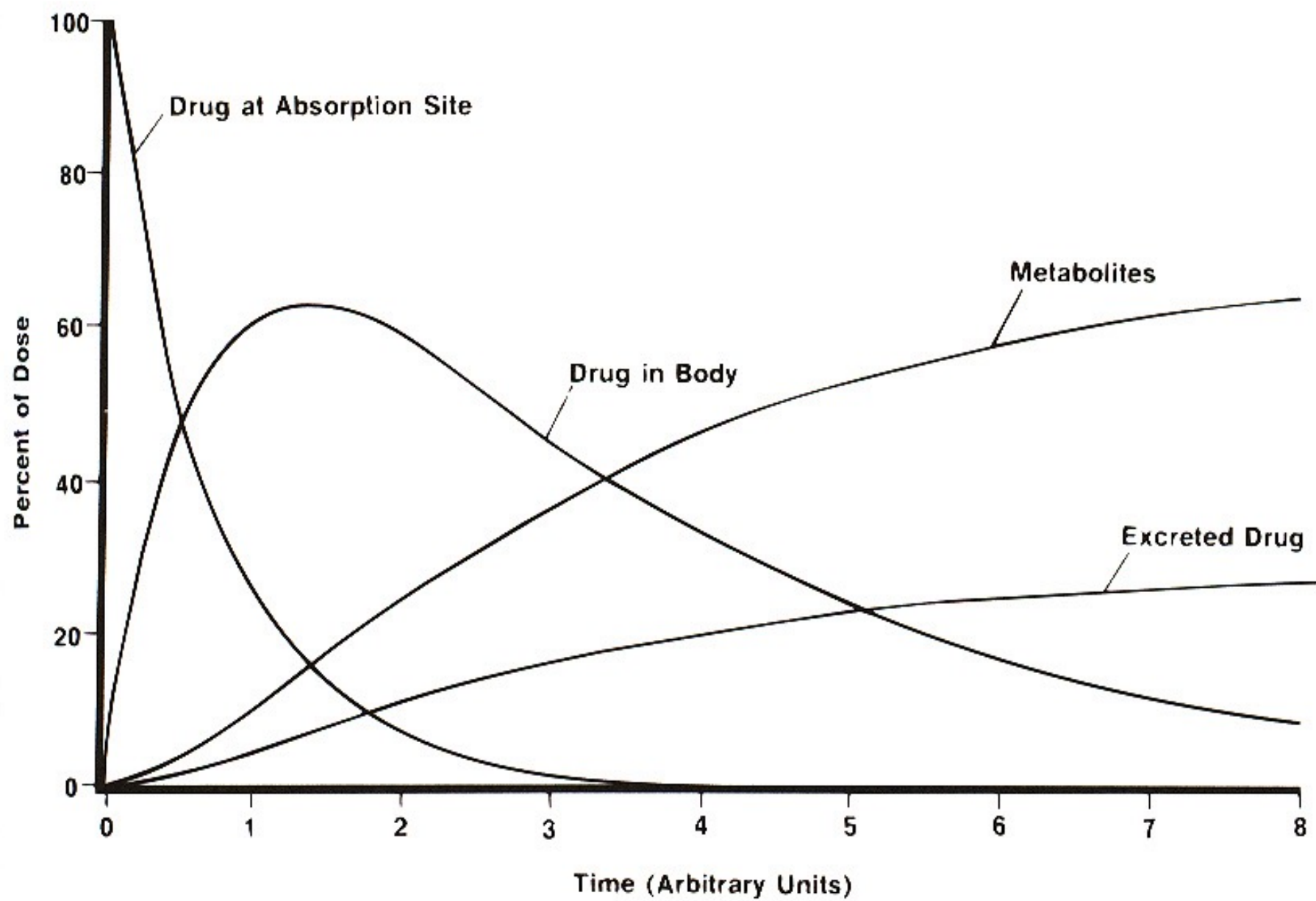


Fig. 3-5. Time course of drug in the body. (From Rowland, M., and Tozer, T.N.: Clinical Pharmacokinetics. 2nd Ed., Philadelphia, Lea & Febiger, 1989.)

Elimination of drugs from the body

MAJOR

KIDNEY

filtration
secretion
(reabsorption)

LIVER

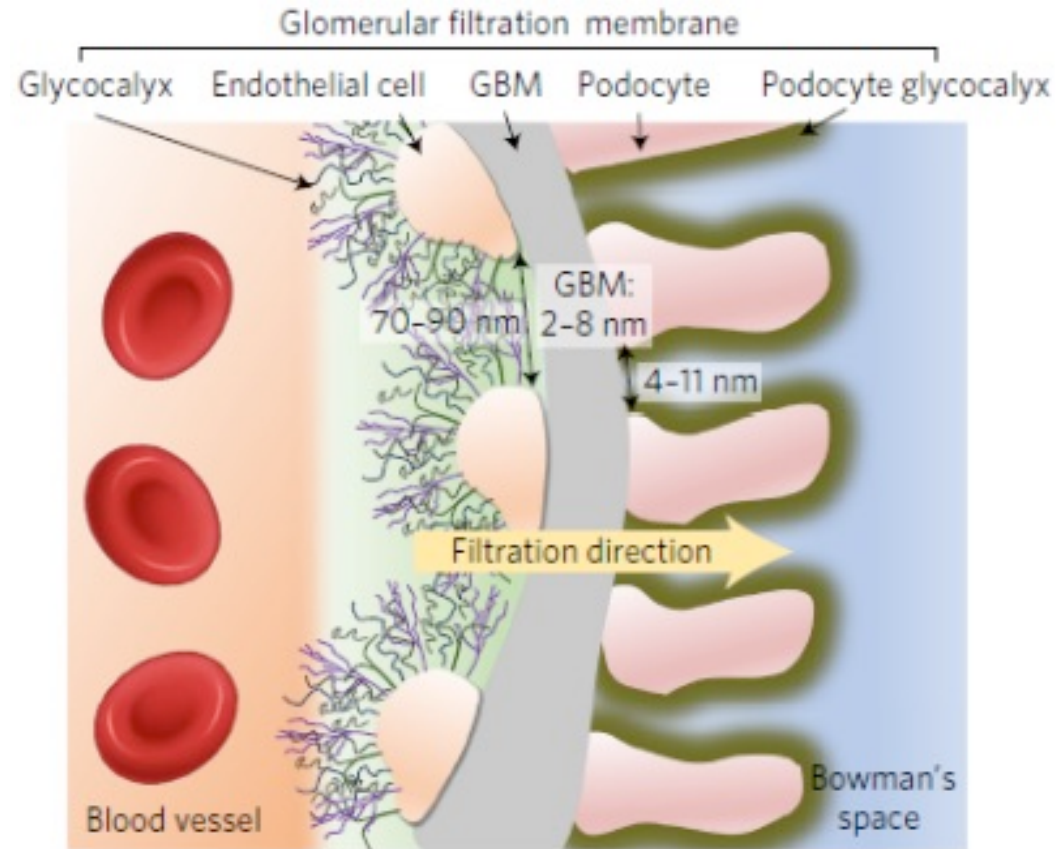
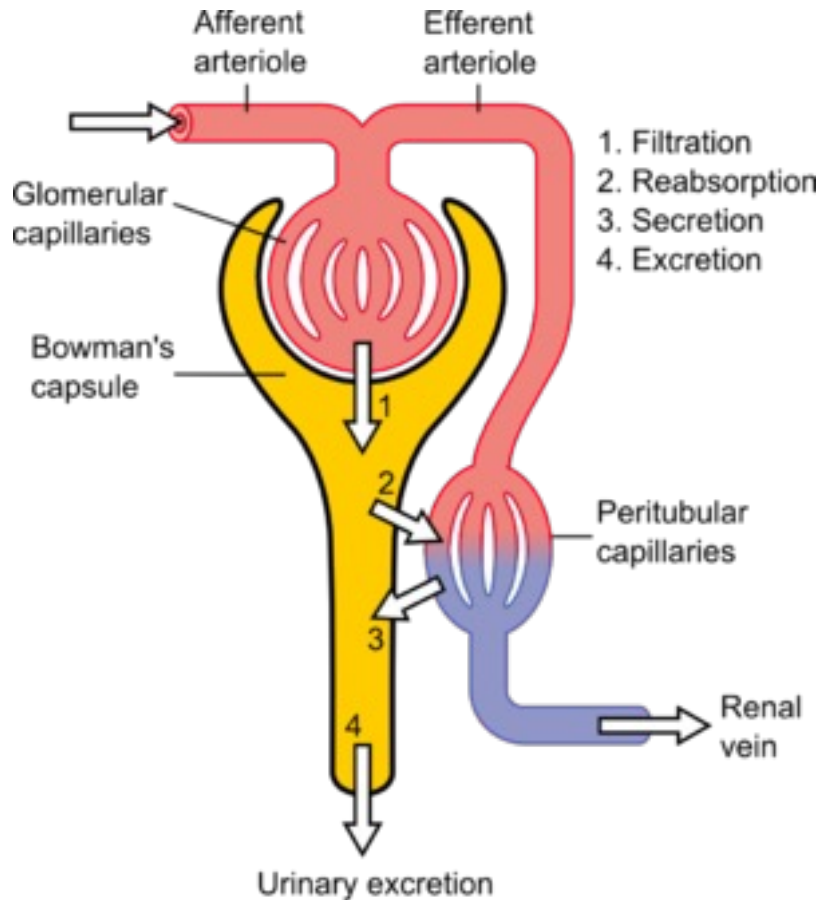
metabolism
secretion

MINOR

LUNGS
exhalation

OTHERS
mother's milk
sweat, saliva etc.

Elimination by the Kidney



glomerular filtration membrane (GBM)

Particles above 10nm can't be excreted by kidney

Elimination

- **Zero order:** constant rate of elimination irrespective of plasma concentration.
- **First order:** rate of elimination proportional to plasma concentration. Constant *Fraction* of drug eliminated per unit time.

Rate of elimination $\propto$ Amount

Rate of elimination = $K \times$ Amount

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Zero Order Elimination: Ethanol

- ▶ Ethanol is distributed in total body water.
- ▶ Mild intoxication: 1 mg/ml in plasma.
- ▶ How much should be ingested to reach it?

Answer: 50L of water in our body so, 50 g or 63 ml of pure ethanol

130 ml of a strong alcoholic drink like whiskey (2 large peg!)

- ▶ Ethanol has a constant elimination rate = **10 ml/h**
- ▶ To maintain mild intoxication, at what rate must ethanol be taken now?

at 10 ml/h of pure ethanol, or 20 ml/h of drink (less than 1 small peg!!).

DRUNKENNESS → Coma → Death

First Order Elimination

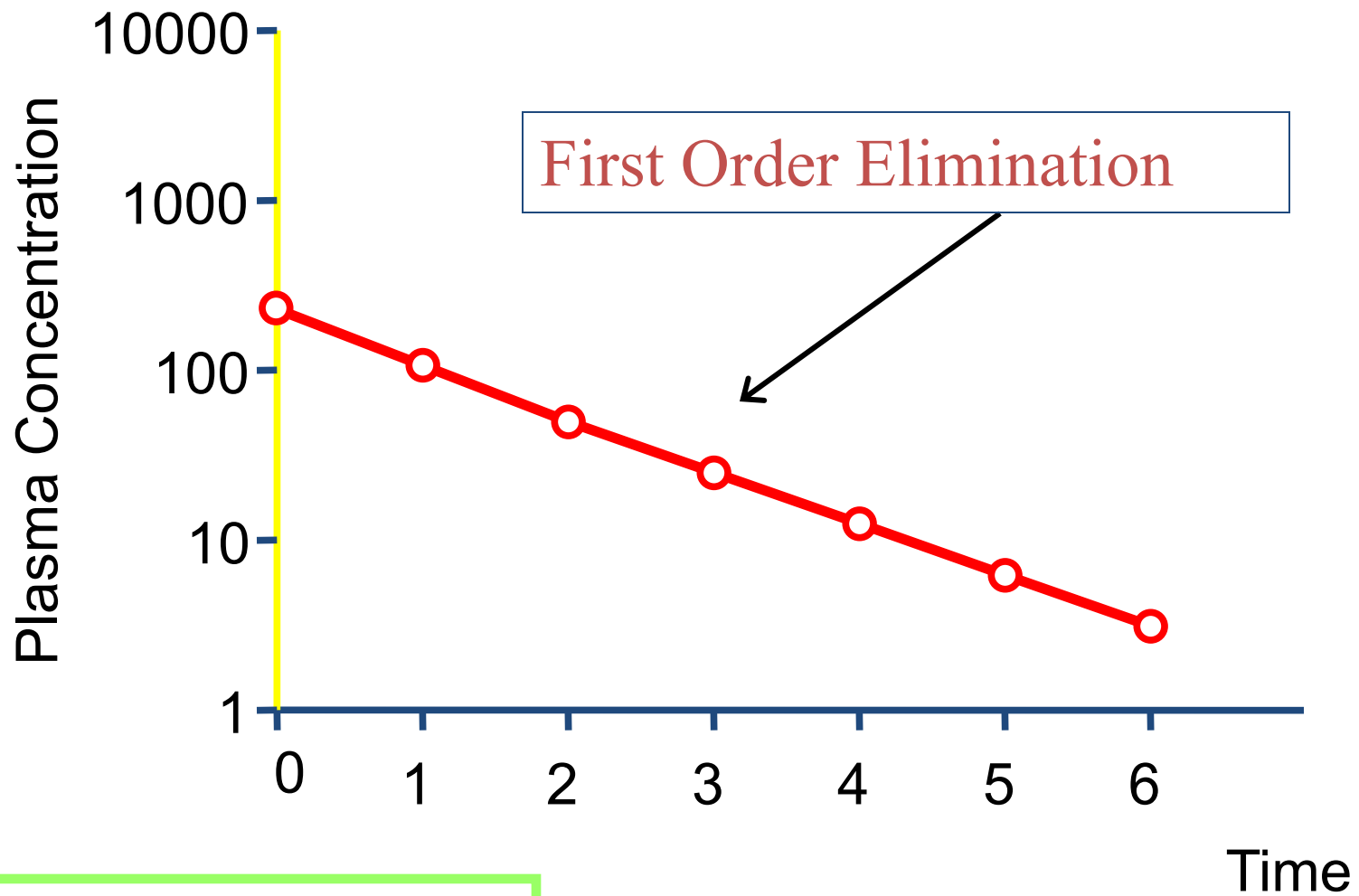
- **Clearance**: volume of plasma cleared of drug per unit time.

Clearance = Rate of elimination ÷ plasma conc.

- **Half-life of elimination**: time for plasma conc. to decrease by half.

Useful in estimating:

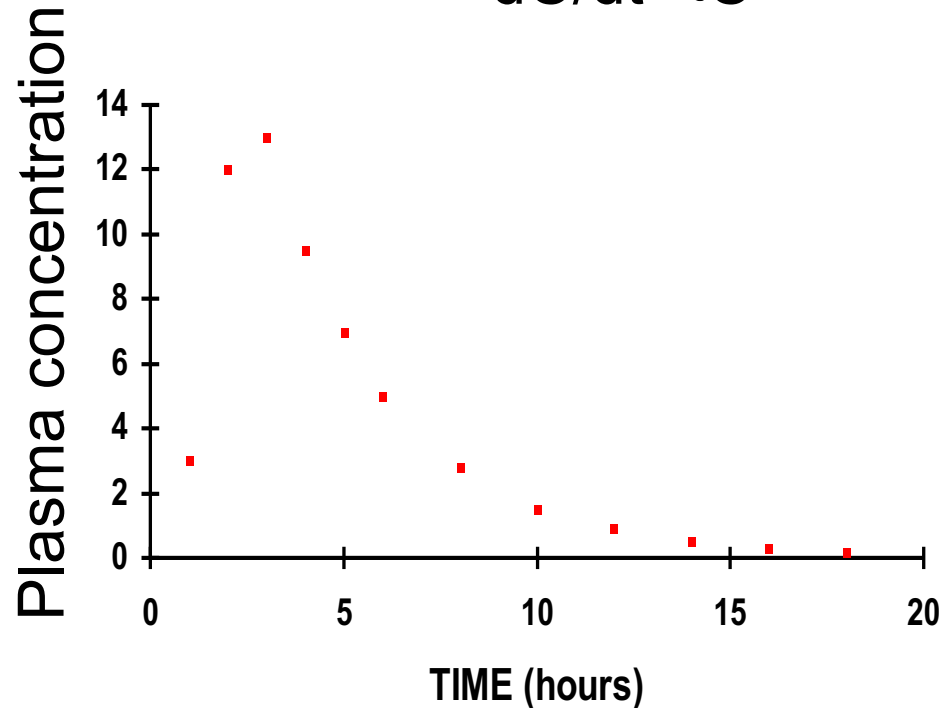
- time to reach steady state concentration.
- time for plasma concentration to fall after dosing is stopped.



$$\log C_t = \log C_0 - \frac{K_{el} \cdot t}{2.303}$$

First Order Elimination

$$dC/dt \propto C$$



$$dC/dt = -k \cdot C$$

$$C_t = C_0 \cdot e^{-K_{el} \cdot t}$$

$$\ln C_t = \ln C_0 - K_{el} \cdot t$$

$$y = b - a \cdot x$$

Biological half-life

Elimination of drugs from the body usually follows first order kinetics with a characteristic half-life ($t_{1/2}$) and fractional rate constant (K_{el}).

An Example – How long will it take for the body to eliminate 100 mg of drug X?

- 100mg of drug in plasma @ time zero and the drug's half life is 4 hours
- 50% of drug is eliminated in one half life
 - 50mg of drug remains in plasma after 4 hours (1 half life)
 - 25mg of drug remains in plasma after 8 hours (2 half lives)
 - 12.5mg of drug remains in plasma after 12 hours (3 half lives)
 - 6.25mg of drug remains in plasma after 16 hours (4 half lives)
- Thus for drugs that are eliminated via first order kinetics >90% of a drug would be eliminated in 4 half lives

Pharmacokinetics: Single compartment model



Entire body act as a single, uniform compartment

Concentration in the plasma and tissues is considered to be uniform

Drug considered to be distributed in single compartment

Pharmacokinetics: Two compartment model

Body divided into central and peripheral compartment

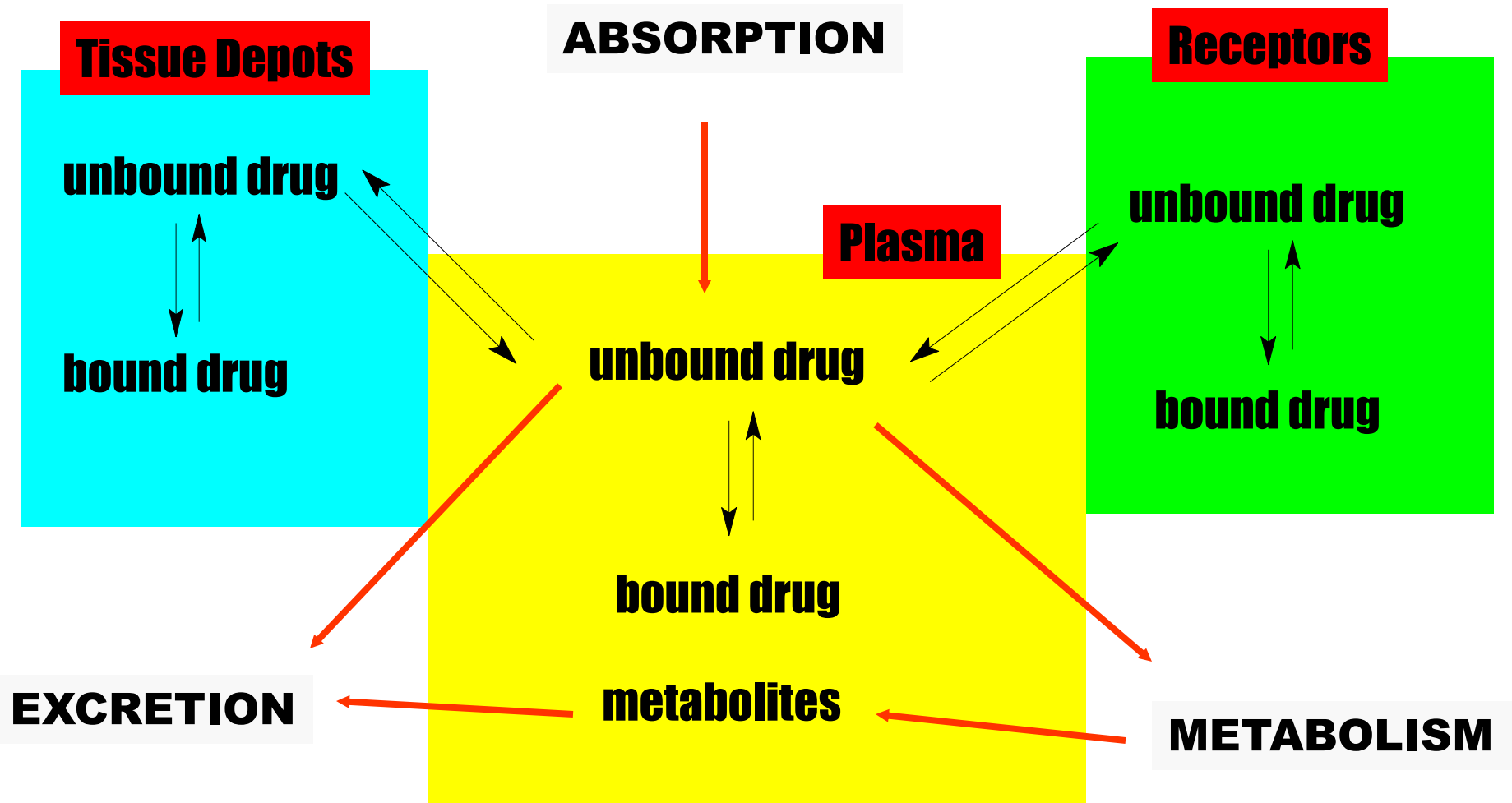
Central compartment: Blood and highly perfused tissues (Kidney, heart, brain etc.)

Peripheral compartment: Rest of tissues (adipose tissue, bone, teeth)

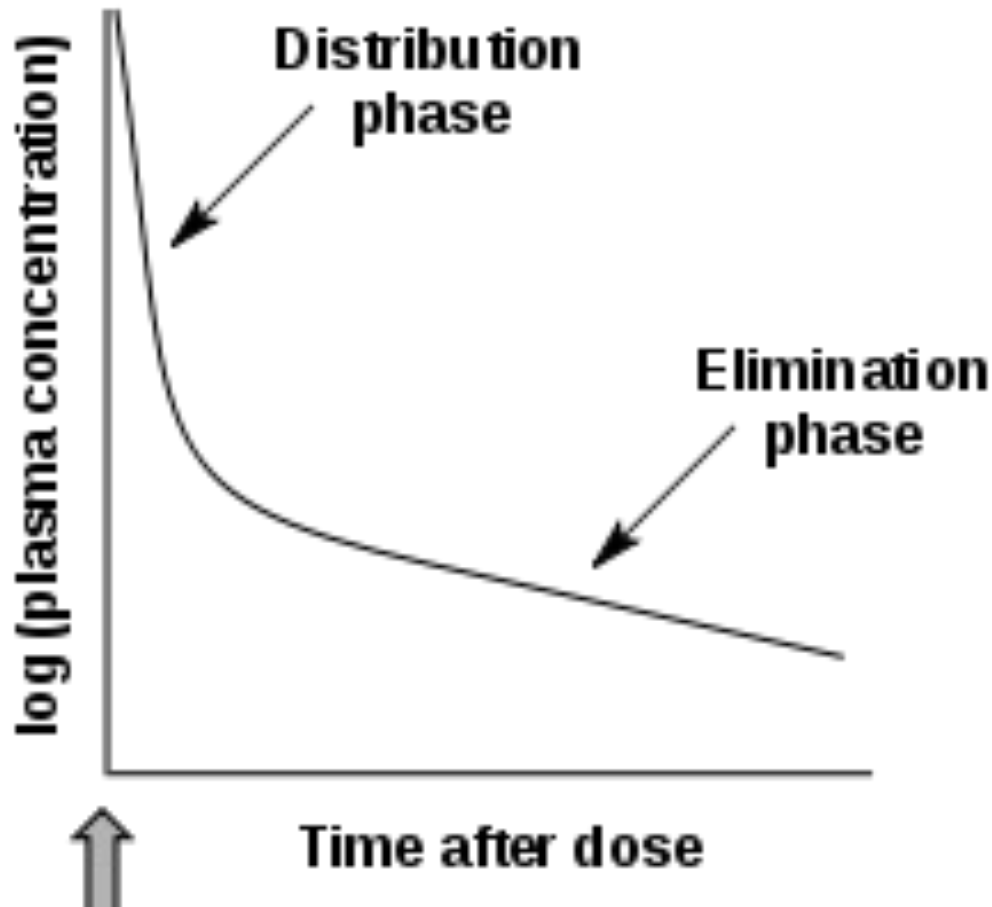
Central and peripheral also have a first order distribution process

Plasma drug concentration declines bi-exponentially

Pharmacokinetics: Two compartment model



Pharmacokinetics: Two compartment model



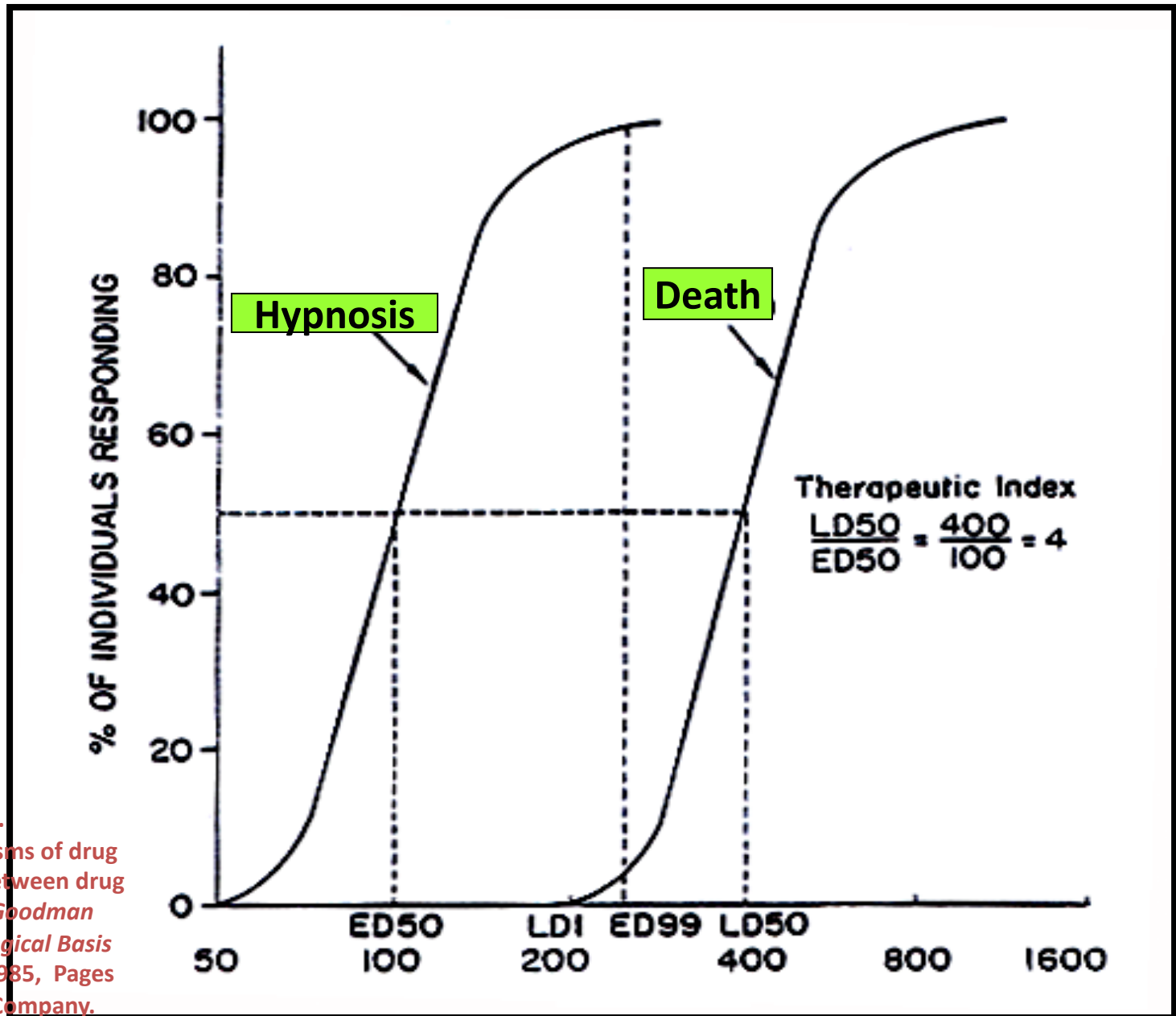
Effectiveness, toxicity, lethality

- ED50 - Median Effective Dose 50; the dose at which 50 percent of the population or sample manifests a given effect
- TD50 - Median Toxic Dose 50 - dose at which 50 percent of the population manifests a given toxic effect
- LD50 - Median Lethal Dose 50 - dose which kills 50 percent of the subjects

Quantification of drug safety

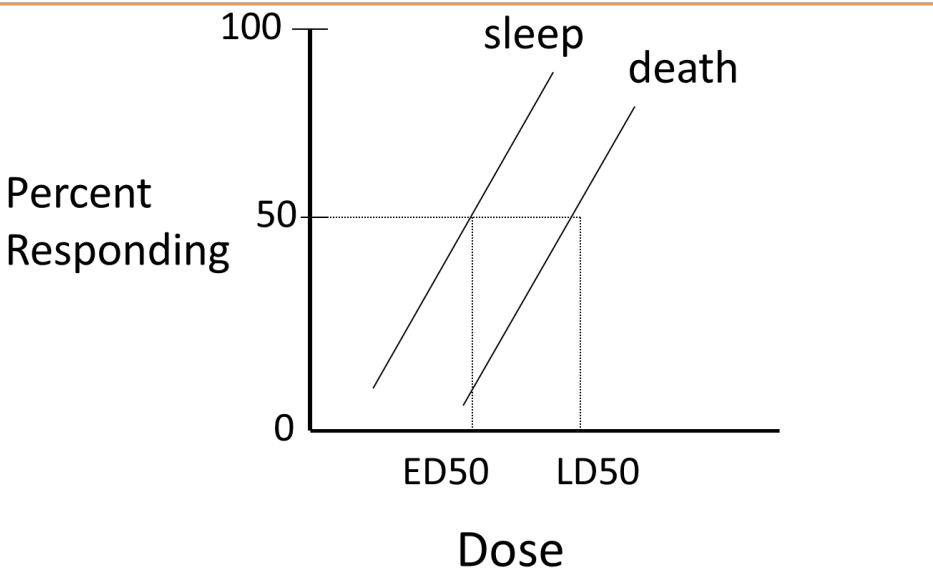
$$\text{Therapeutic Index} = \frac{\text{LD50}}{\text{ED50}}$$

THERAPEUTIC INDEX – AN INDEX OF SAFETY

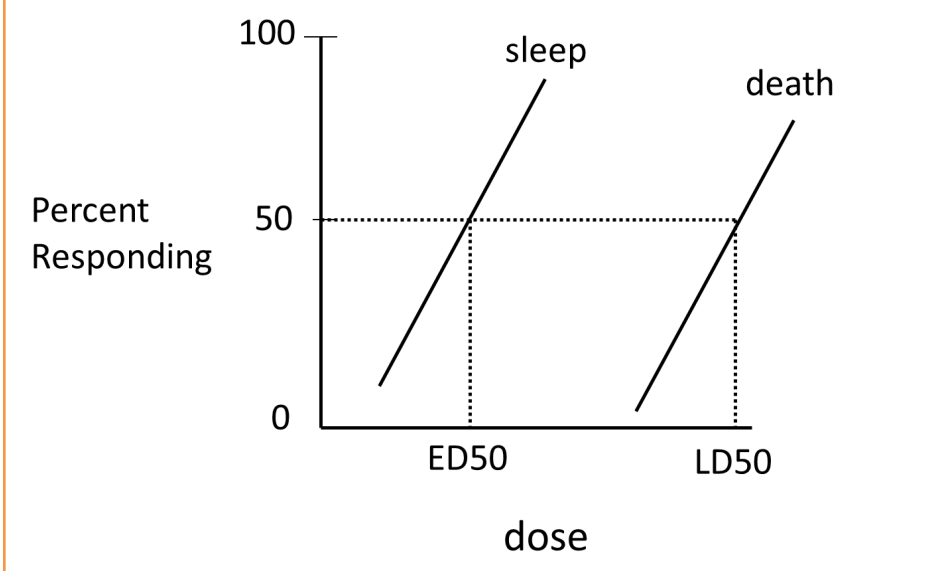


From Ross EM and Gilman AG.
Pharmacodynamics: mechanisms of drug
action and the relationship between drug
concentration and effect.. in *Goodman
and Gilman's The Pharmacological Basis
of Therapeutics*. 7th edition, 1985, Pages
35-48. Macmillan Publishing Company.

Drug A

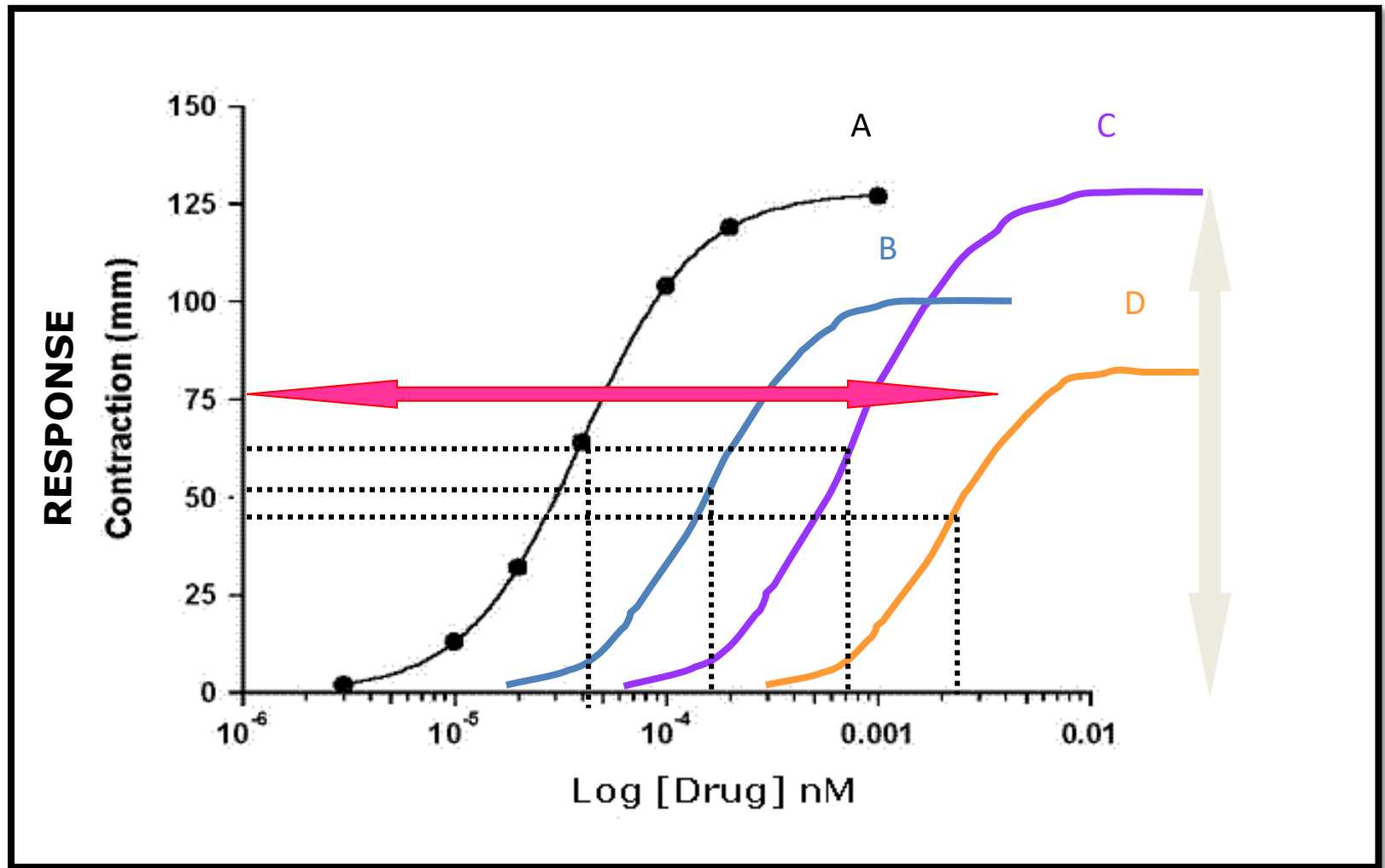


Drug B



Which is better?

Exercise



RANK ORDER OF POTENCY: $A > B > C > D$

RANK ORDER OF EFFICACY OF RESPONSE: $A = C > B > D$

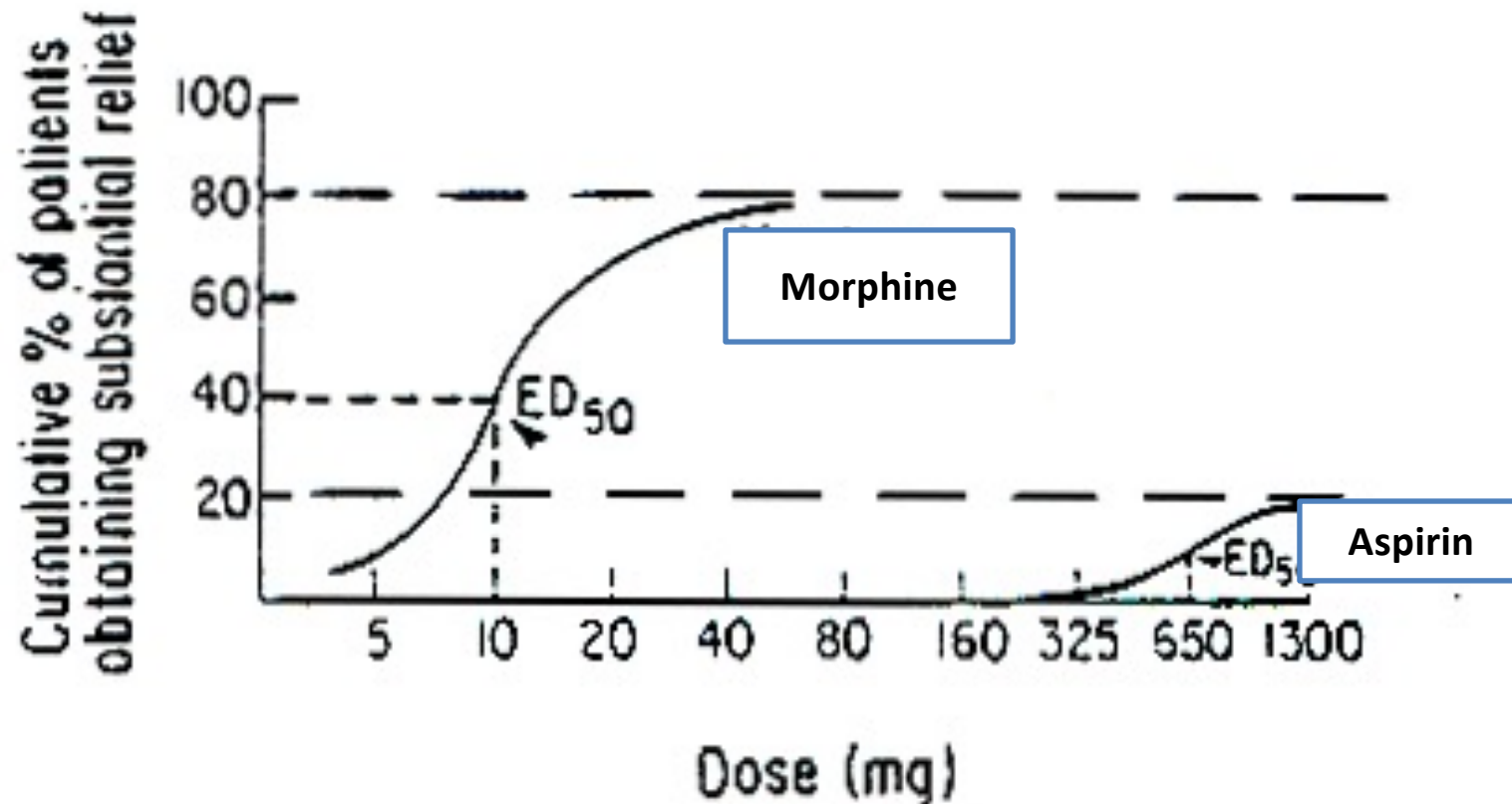
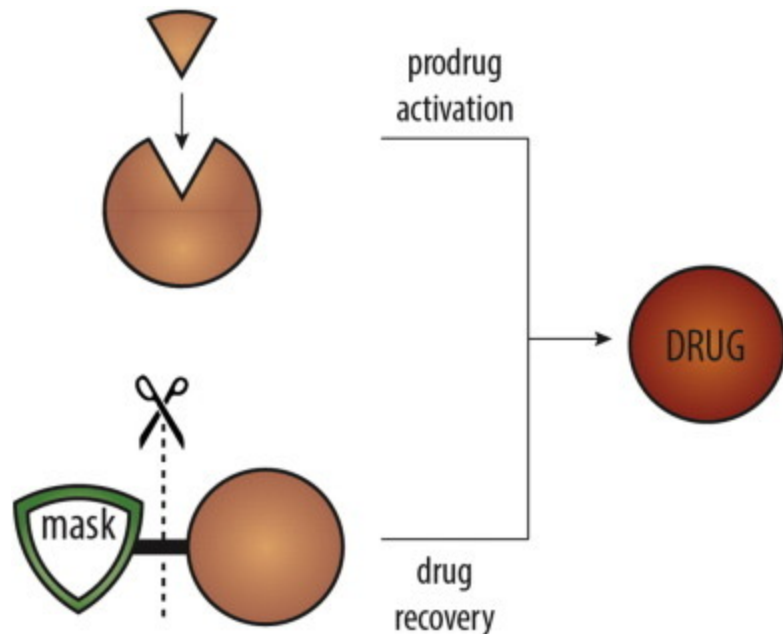


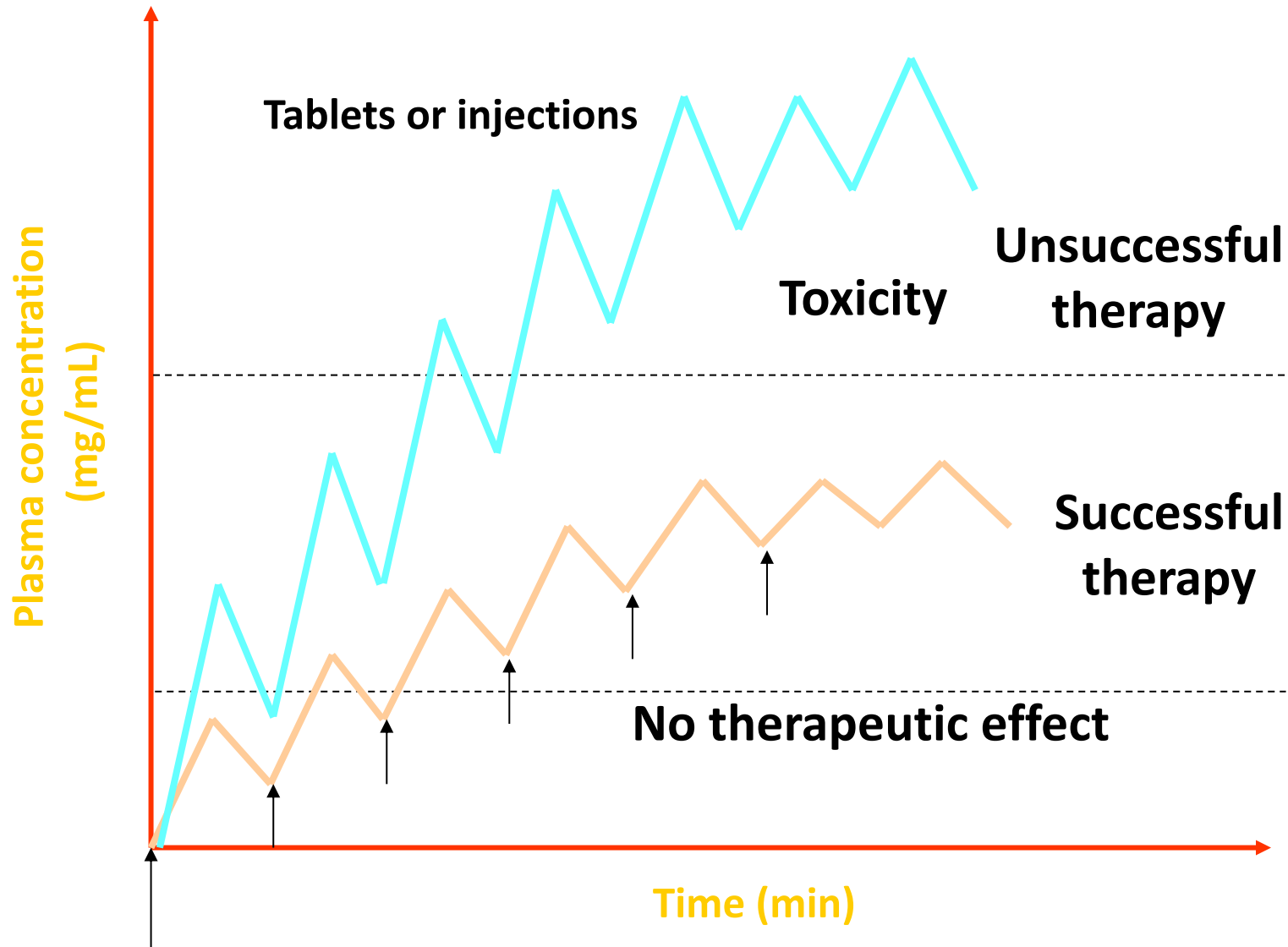
Fig. 1-5. Dose-response curves describing the ability of aspirin and morphine to relieve pain in patients with migraine headache. Note that morphine is both more potent (has a lower ED₅₀), and has a greater overall efficacy (ceiling effect).

Prodrugs

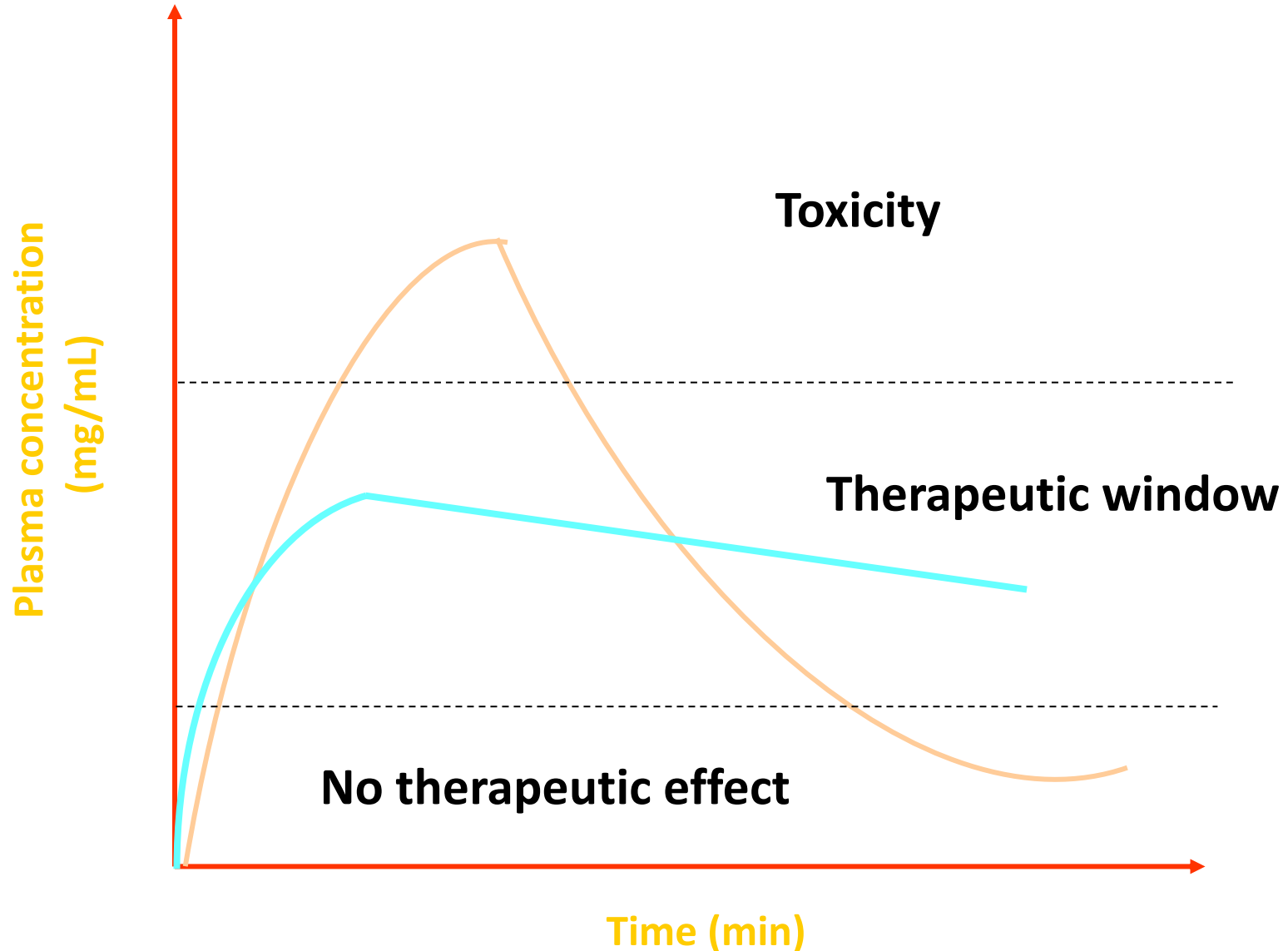
- Compounds that get metabolized into active drug
- Prodrugs by themselves have low or no activity
- May result in increasing bioavailability if the active drug is not well soluble or absorbed
- Improves targeting as the signal (enzymes) to convert the pro-drug to active drug may be localized to target tissue
- 10% of all marketed drugs are prodrugs



Plasma Concentration – Multiple Administrations



Plasma Concentration – Single Administration



Controlled release and long circulation: Advantages

- **Increase patient compliance**
- **Employ less total drug**
- **Improve efficiency in treatment**

Biomaterials for controlled release

- **External vs Internal**
 - External devices such as pumps, skin patches
 - Lot more flexibility with external material
 - Internal is qualified as “Biomaterial” meaning it is safe and would not cause side effects
- **Biomaterials**
 - Any material that is considered safe for human use
 - Could be synthetic (e.g. synthetic polymer) or natural (e.g. alginate)
 - Could be metallic (e.g. gold, silver), ceramic, polymeric etc.